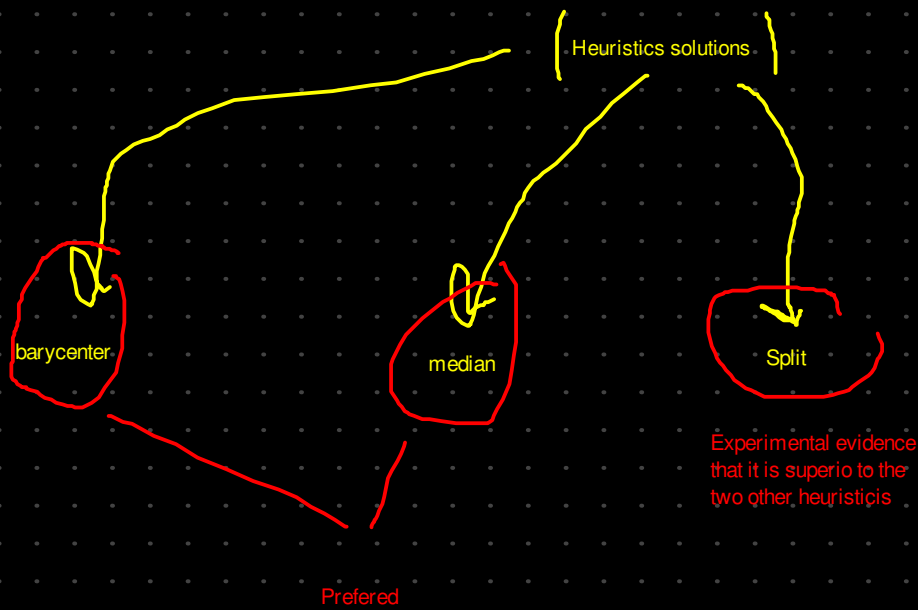


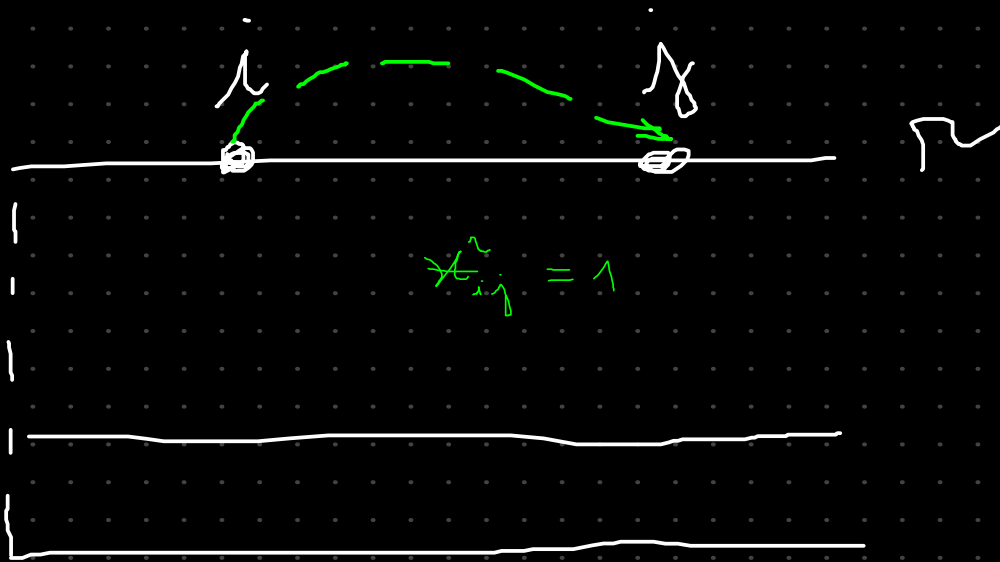
Solve multi-level crossing minimization problem



One-level crossing minimisation problems



ILP representation of the multi-level crossing minimisation problem



Minimizing number crossings

$$\sum_{n=1}^{p-1}$$

level

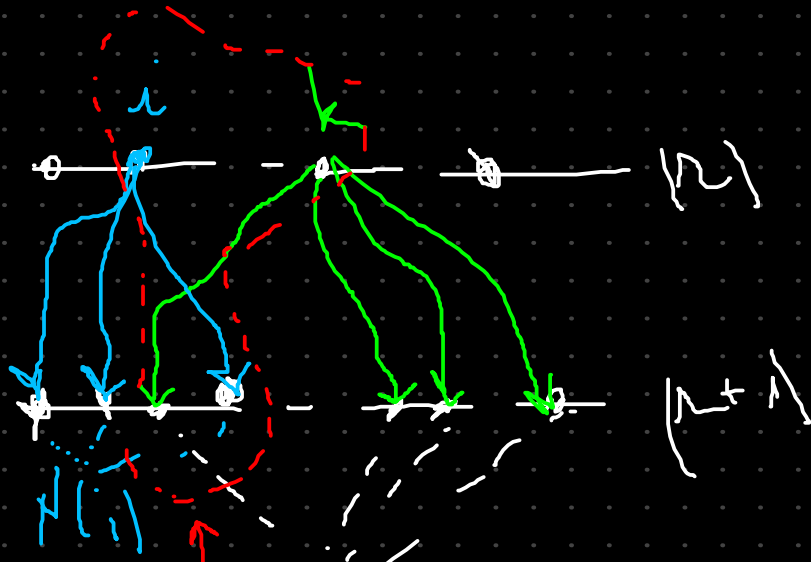
$$\sum_{i=1}^{|V_n|-1}$$

fix v

$$\sum_{j \in N(i)}$$

$$\sum_{k \in N(k)}$$

all unordered pairs



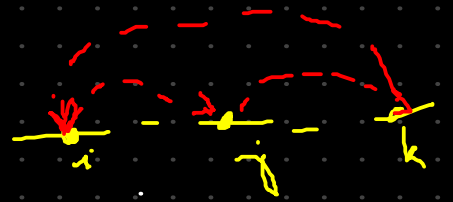
$$x_{ik}^n x_{jl}^{n+1} + x_{ki}^n x_{jl}^{n+1}$$

$\rightarrow 1 \text{ if } i \parallel j$

We are adding this part because if k is before i then the first part will result in 0 even tho we might have a crossing between edges (i,j) and (l,j)

subject to: avoid 3-cycles

why just 3-cycles?



$$\begin{cases} (1) & x_{ij}^{\wedge} + x_{jk}^{\wedge} + x_{ki}^{\wedge} \leq 2, \quad 1 \leq i < j < k \leq |V_n| \\ (2) & x_{ij}^{\wedge} + x_{ji}^{\wedge} = 1, \quad 1 \leq i < j \leq |V_n|, x_{ij}^{\wedge} \in \{0, 1\} \end{cases}$$

(3) feasible order

Quadratic integer objective

Difficult to solve with the available solvers

Linearisation of the objective function

$$x_{ji}^{\wedge} = 1 - x_{ij}^{\wedge} \quad (3) \quad , \quad c_{ijkl}^{\wedge} = \begin{cases} 1 & (i,j) \& (k,l) \text{ con} \\ 0 & \text{otherwise} \end{cases}$$